

Ethan Sotka

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EDUCATION

University of Notre Dame | **B.S. in Computer Science, Applied Computational Mathematics and Statistics**

May 2027

Honors: 4x Dean's List, President's Circle Mentor Program, Visa FinTech Foundations

GPA: 3.9

TECHNICAL AND LANGUAGE SKILLS

Technical: Python(*Advanced*), SQL(*Advanced*), C(*Intermediate*), Excel(*Intermediate*), R(*Intermediate*), Git(*Intermediate*), C++(*Basic*), Kotlin(*Basic*)

Frameworks: NumPy (*Intermediate*), Pandas (*Intermediate*), Matplotlib/Seaborn (*Intermediate*), BeautifulSoup (*Basic*), TensorFlow (*Basic*)

WORK EXPERIENCE

FTI Consulting | McLean, VA

May 2026 - Present

Consulting Intern

- Analyzed **SQL** datasets of **500M+ records** on telecom network, market, and financial data, in conjunction with **Excel** transformations to translate technical findings into client-ready insights, informing **hundreds of millions of dollars in asset backed loans** for clients
- Collaborated across teams to integrate **QGIS** maps and analyses into executive-level reports, supporting **multibillion-dollar divestiture**

Exiger | McLean, VA

June 2025 - Jan 2026

Data Analyst (Part-Time)

- Continued to develop app adding **API** calls and **Python** integration to Snowflake, allowing pipeline to update through single input file

Data Analyst Summer Intern

- Built and automated **SQL** workflows to analyze supplier part datasets, accurately identifying underlying mineral and elemental compositions; streamlined internal processes to **save ~40 hours/week** and directly supported fulfillment of **\$10M+** in client contracts
- Supplemented **3M+** supply chain records using **Python (pandas)**, enhancing data quality and supporting scalable operations

Lucy Family Institute for Data & Society | Notre Dame, IN

Jan 2025 - Dec 2025

DM2 Lab Research Assistant

- Engineered an automated pipeline to parse research papers and datasets, streamlining the validation of scientific reproducibility
- Utilized **ChatGPT API Agent** to generate Python scripts that reproduce study figures with **92% accuracy** from raw data inputs

DLAL Lab Research Assistant

- Developed **Python** program that leveraged **ChatGPT API** with knowledge graph as a **RAG** to test if LLMs hold core beliefs
- Increased response accuracy by **24%** through testing simple feeder questions in chain of response trials, optimizing response quality

PROJECTS

Blackjack Card Counting Trainer

Jan 2026 - Present

- Designed multi-module **Python** application leveraging object-oriented programming, event-driven architecture, and graphical user interface development with **Pygame** allowing players to experience diverse blackjack situations while learning advantage play
- Built **Monte Carlo** simulation extension to explore card-counting algorithms, modeling blackjack profitability outcomes with **matplotlib**

Sports Betting Arbitrage Tool

March 2024 - Present

- Utilized **Selenium**, **BeautifulSoup**, and **Pandas** in **Python** to repeatedly scrape betting odds from several sports betting websites
- Applied Positive Expected Value and Arbitrage strategies with Kelly Criterion to data in simulation to create positive returns

Predicting Longevity/Success at QB in the NFL

August 2024 - Dec 2024

- Applied multiple Linear Regression with tests for assumptions to evaluate college football statistics using **R**, to develop predictive models for projecting NFL quarterback win rates and discovering no predictive power for longevity in the NFL
- Submitted research paper to **Baltimore Ravens**, with positive feedback from Senior Manager and invitation for further research

LEADERSHIP AND ACTIVITIES

Quant Club of Notre Dame | Notre Dame, IN

August 2023 - Present

Officer

- Applied Game Theory, Probability, and Risk Management in real-time simulations through strategy games and competitions
- Explored quantitative finance best practices through lectures involving professional Quants to understand Quantitative trading

DELTA (Discovering Econometrics: Learning Through Application) | Notre Dame, IN

August 2024 - Present

Member / Speaker

- Applied mathematical and statistical methods to solve real-world problems across various disciplines in short time frames
- Developed an advanced understanding of quantitative tools used by economists and data scientists through professional oversight

DataFest 2024, 2025 | Notre Dame, IN

Finalist

March 2024 & March 2025

- Analyzed several data sets of 700,000+ points with **numPy** and **pandas** for CourseKata to improve user learning and enjoyment
- Presented trends and patterns in data to CourseKata along with actionable next steps to allow advancement of teaching practices

Interests: Football, Entrepreneurship, Neuroscience, Fitness, Coding, Jiu Jitsu, Investing, Baltimore Ravens, Strategic/Number Games